Coupling Integrals of Polynomial Cubes at a Welded Elastic Interface

The Static Image in Closed Form, and the Plane Waves for the Rest

T. M. Nestor

Independent researcher, Australia

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Abstract

In a stratified elastic reference medium the Green's tensor is that of a homogeneous medium plus the waves reflected and transmitted at the interfaces, and it is most naturally available not as a tabulated function but through its plane-wave, up- and down-going, factorisation. A volume-integral method that divides the heterogeneity into cubic cells carrying polynomial fields needs the Galerkin coupling of every pair of cells through this reference. Between cells in planes that do not touch, the plane-wave representation with the cells' moments in closed form is exact, and its integral over the lateral wavenumber converges exponentially. Next to a welded interface it does not: a cell is coupled to itself and to its neighbours through its own mirror image, which touches them, and the reflected coupling converges only algebraically. We show that the singular part is the static field of a point force at the interface, to which the reflected and transmitted kernels tend as $(\omega/\beta q)^2$ at large lateral wavenumber q, and obtain it and its coupling blocks in closed form. Solved in the lateral-wavenumber domain, Rongved's problem gives every entry as $\mathrm{e}^{-q\zeta}/q$ times a polynomial; in space the reflected and transmitted static fields are 356 and 275 terms built on Kelvin's potential and two logarithmic potentials, Mindlin's terms, which at a welded interface are present for every pair of distinct media and which we write as a line of images. The moments of these kernels over a box with the singular point at a vertex reduce, by Euler's identity for homogeneous functions, to faces, edges and edges to infinity, with Hadamard finite parts where the degree is negative and two base integrals where it is zero, $B_1 = \frac{5}{4}\mathrm{Cl}_2(\pi/3) - G$ and $B_2 = \frac{1}{3}\log[(1+\sqrt{2})^2/(2(1+\sqrt{3}))]$, so that every moment is elementary apart from Catalan's constant and one value of Clausen's function. The corner moments and the blocks agree with independent routes to 10^{-15} and 10^{-13} or better, coincide with the blocks of the homogeneous medium to 2 to 8×10^{-15} when the two media are the same, and with the static part subtracted the plane-wave remainder converges as Q^{-4} in the cutoff Q instead of Q^{-2}. The next term of the kernels, of order ω^2 with a coefficient that does not depend on the frequency, is derived in the wavenumber domain; subtracted as well, it leaves a remainder that converges as Q^{-6}, so that a double-precision plane-wave integral truncated at $Qh = 10$ gives the block of a touching cell to 4×10^{-10} at $\omega h/\beta = 1/6$.

1 Introduction

A volume-integral method for elastic waves represents a heterogeneous body as a contrast with a reference medium, divides the contrast into cells, and couples every pair of cells through the Green's tensor of the reference. When the reference is homogeneous the coupling of cubic cells carrying polynomial fields can be obtained in closed form and evaluated to round-off [1]. In seismology the natural reference is stratified, and its Green's tensor is that of a homogeneous medium plus waves reflected and transmitted at the interfaces. The stratified Green's tensor is rarely tabulated: it is computed through its plane-wave representation, in which, at each lateral wavenumber, waves travel up and down through the layers and are reflected and transmitted at each interface [2, 3]. A volume method built on a stratified reference then needs the coupling blocks in a form that uses only this up- and down-going factorisation [4]; tabulating the Green's tensor between all pairs of depths would cost storage quadratic in the number of depth planes, where the plane waves need only linear storage.

For cells in different planes that do not touch, the plane-wave representation with the cells' moments in closed form is exact, and its integral over the lateral wavenumber converges exponentially because the evanescent waves decay between the planes. For cells in touching planes it diverges, and those blocks are the closed forms of the homogeneous medium [1]. At an interface the same failure reaches further. The reflection of a cell's field is the field of its mirror image, and the image of a cell that touches the interface touches the cell itself and its neighbours along the interface. The reflected coupling of every such cell is therefore singular, and its plane-wave integral converges only algebraically.

This paper removes that obstacle. Its contributions are the following.

1. The plane-wave representation of the coupling blocks, stated for the direct, reflected and transmitted couplings with the cells' moments in closed form (Section 2).
2. The identification of the singular part of the reflected and transmitted couplings as the static field of a point force at a welded interface (Rongved's problem), with the dynamic kernels tending to it as $(\omega/\beta q)^2$; subtracting it changes the convergence of the plane-wave remainder from Q^{-2} to Q^{-4} (Sections 2 and 3).
3. The static image and the transmitted static field in space, as 356 and 275 canonical terms on three potentials, with Mindlin's logarithmic terms written as a line of images (Section 3).
4. The coupling blocks of these fields in closed form: the moments over a box with the singular point at a vertex reduce to elementary functions and two constants, B_1 and B_2, which we evaluate in closed form (Sections 4 and 5).
5. The term of order ω^2 of the reflected and transmitted kernels, by perturbation of the static problem, and its subtraction in the wavenumber domain with a frequency-independent coefficient; the remainder then converges as Q^{-6}, and a cutoff low enough for double-precision mode sums suffices (Section 6).
6. Verification of each step against routes that share no step with it (Section 7).

The dynamic remainder, smooth by construction, is left to the plane waves; Section 8 describes how they carry it through a stack of layers.

2 Cells, blocks and the plane-wave representation

Cells and blocks. A cell is a cube of half-width h centred at $\mathbf{x}_c$, with local coordinates $\boldsymbol{\xi} = (\mathbf{x}-\mathbf{x}_c)/h \in [-1,1]^3$. It carries a nine-component state, the displacement and an independent engineering strain, as a polynomial in $\boldsymbol{\xi}$ [5]; the field is tested against products of Legendre polynomials $L_a(\boldsymbol{\xi}) = \prod_i P_{l_{a,i}}(\xi_i)$, and the source density, the contrast times the state, is written in the monomials $m_c(\boldsymbol{\xi}') = \prod_i \xi_i'^{e_{c,i}}$. With $\mathbf{P}(\mathbf{x},\mathbf{x}')$ the 9×9 point propagator of the reference medium, which maps a point source of force and engineering stress at $\mathbf{x}'$ to the displacement and engineering strain at $\mathbf{x}$, a field cell V_m and a source cell V_n couple through the blocks

$$\mathbf{K}_{ac} = \int_{V_m}\int_{V_n} L_a(\boldsymbol{\xi})\,\mathbf{P}(\mathbf{x},\mathbf{x}')\,m_c(\boldsymbol{\xi}')\,\mathrm{d}^3x\,\mathrm{d}^3x'. \tag{1}$$

In a homogeneous medium $\mathbf{P}$ depends on $\mathbf{x}-\mathbf{x}'$ only, and with $\mathbf{x} = \mathbf{x}_m + h\boldsymbol{\xi}$, $\mathbf{x}' = \mathbf{x}_n + h\boldsymbol{\xi}'$ the block becomes an integral over the separation $\mathbf{s} = h(\boldsymbol{\xi}-\boldsymbol{\xi}') \in [-2h,2h]^3$ against the weight $W_{ac}(\mathbf{s}) = \prod_i w_{l_{a,i}e_{c,i}}(s_i)$, with $w_{\alpha\beta}(s) = h\int v^\alpha (v - s/h)^\beta\,\mathrm{d}v$ the cross-correlation of the two cells' one-dimensional factors, a piecewise polynomial on eight boxes [1]. The kernel is singular at $\mathbf{s} = -(\mathbf{x}_m - \mathbf{x}_n)$, which for touching cells is a vertex of some of the boxes.

The plane-wave representation. Take coordinates in the order (z,x,y), z the depth, and a lateral wavenumber $\mathbf{k} = (k_x,k_y) \in \mathbb{R}^2$, $q = |\mathbf{k}|$. For each wave $\nu \in \{\mathrm{P},\mathrm{SV},\mathrm{SH}\}$ of speed c_ν ($c_\mathrm{P} = \alpha$, $c_\mathrm{SV} = c_\mathrm{SH} = \beta$) let

$$k_{z,\nu} = \left(\omega^2/c_\nu^2 - q^2\right)^{1/2}, \quad \operatorname{Im} k_{z,\nu} \geq 0, \qquad \mathbf{k}_\nu^\sigma = (\sigma k_{z,\nu}, k_x, k_y), \quad \sigma = \pm 1,$$

the wave vector travelling down ($\sigma = +1$) or up ($\sigma = -1$). Its polarisations are

$$\mathbf{e}_\mathrm{P}^\sigma = \frac{\alpha}{\omega}\,\mathbf{k}_\mathrm{P}^\sigma, \qquad \mathbf{e}_\mathrm{SH} = \frac{(0,-k_y,k_x)}{q}, \qquad \mathbf{e}_\mathrm{SV}^\sigma = \frac{\beta}{\omega}\,\mathbf{k}_\mathrm{SV}^\sigma \times \mathbf{e}_\mathrm{SH},$$

with $\mathbf{e}_\mathrm{SH} = (0,1,0)$ at $\mathbf{k} = \mathbf{0}$. Each is a unit vector in the bilinear sense, $\mathbf{e}\cdot\mathbf{e} = 1$, which, unlike $\mathbf{e}\cdot\bar{\mathbf{e}} = 1$, continues analytically into the evanescent range $q > \omega/c_\nu$. The state of the wave is the nine-vector $\mathbf{d}_\nu^\sigma = (\mathbf{e},\boldsymbol{\varepsilon})$ of its displacement and engineering strain, $\varepsilon_{ij} = \frac{\mathrm{i}}{2}(k_i e_j + k_j e_i)$ with $\mathbf{k} = \mathbf{k}_\nu^\sigma$ and the three shears doubled, and $\mathbf{M} = \operatorname{diag}(1,1,1,1,1,1,\frac{1}{2},\frac{1}{2},\frac{1}{2})$ pairs the engineering-stress entries of a source with the engineering-strain entries of a state. In a homogeneous medium of density ρ, for $\sigma(z-z') > 0$,

$$\mathbf{P}(\mathbf{x},\mathbf{x}') = \frac{1}{4\pi^2}\int_{\mathbb{R}^2}\sum_\nu w_\nu\,\mathbf{d}_\nu^\sigma\,\mathbf{d}_\nu^{\sigma\mathsf{T}}\mathbf{M}\,\mathrm{e}^{\mathrm{i}\mathbf{k}_\nu^\sigma\cdot(\mathbf{x}-\mathbf{x}')}\,\mathrm{d}^2k, \qquad w_\nu = \frac{\mathrm{i}}{2\rho c_\nu^2 k_{z,\nu}}, \tag{2}$$

the source entering through $\mathbf{d}_\nu^{\sigma\mathsf{T}}\mathbf{M}$ and the field through $\mathbf{d}_\nu^{\sigma}$, as reciprocity requires. For cells in planes that do not touch, $|z_m - z_n| \geq 4h$, set $\sigma = \operatorname{sgn}(z_m - z_n)$; Eq. (1) then becomes

$$\mathbf{K}_{ac} = \frac{1}{4\pi^2} \int_{\mathbb{R}^2} \sum_\nu w_\nu \, \Lambda_a(\mathbf{k}_\nu^\sigma) \, \mu_c(-\mathbf{k}_\nu^\sigma) \, \mathbf{d}_\nu^\sigma \, \mathbf{d}_\nu^{\sigma\mathsf{T}} \mathbf{M} \, \mathrm{e}^{\mathrm{i}\mathbf{k}_\nu^\sigma \cdot (\mathbf{x}_m - \mathbf{x}_n)} \, \mathrm{d}^2 k, \tag{3}$$

with the scalar moments of the two cells, both in closed form for complex arguments [1],

$$\Lambda_a(\boldsymbol{\kappa}) = h^3 \int_{[-1,1]^3} L_a(\boldsymbol{\xi}) \, \mathrm{e}^{\mathrm{i}h\boldsymbol{\kappa}\cdot\boldsymbol{\xi}} \, \mathrm{d}^3\xi = h^3 \prod_i 2\,\mathrm{i}^{l_{a,i}} j_{l_{a,i}}(h\kappa_i), \tag{4}$$

$$\mu_c(\boldsymbol{\kappa}) = h^3 \int_{[-1,1]^3} m_c(\boldsymbol{\xi}') \, \mathrm{e}^{\mathrm{i}h\boldsymbol{\kappa}\cdot\boldsymbol{\xi}'} \, \mathrm{d}^3\xi' = h^3 \prod_i F_{e_{c,i}}(h\kappa_i), \qquad F_n(t) = \int_{-1}^{1} \xi^n \mathrm{e}^{\mathrm{i}t\xi} \, \mathrm{d}\xi, \tag{5}$$

j_l the spherical Bessel functions and F_n a finite combination of them.

Reflection and transmission. Let medium A, with Lamé constants λ_A, μ_A and density ρ_A, fill $z < 0$ and medium B fill $z > 0$, welded at $z = 0$ (Figure 1). With the amplitudes referred to the interface and the polarisations of each medium, a down-going wave ν in A gives the up-going wave μ in A with amplitude $R_{\mu\nu}(\mathbf{k})$ and the down-going wave μ in B with amplitude $T_{\mu\nu}(\mathbf{k})$, obtained at each $\mathbf{k}$ from the continuity of displacement and traction, a 6×6 linear system [3]. For $\mathbf{x}$ and $\mathbf{x}'$ both in A the reflected propagator is

$$\mathbf{P}^R(\mathbf{x}, \mathbf{x}') = \frac{1}{4\pi^2} \int_{\mathbb{R}^2} \sum_{\mu,\nu} \mathbf{d}_\mu^{A-} \, R_{\mu\nu} \, w_\nu^A \, \mathbf{d}_\nu^{A+\mathsf{T}} \mathbf{M} \, \mathrm{e}^{\mathrm{i}(\mathbf{k}_\mu^{A-}\cdot\mathbf{x} - \mathbf{k}_\nu^{A+}\cdot\mathbf{x}')} \, \mathrm{d}^2 k, \tag{6}$$

and for $\mathbf{x}'$ in A and $\mathbf{x}$ in B the whole coupling is the transmitted one,

$$\mathbf{P}^T(\mathbf{x}, \mathbf{x}') = \frac{1}{4\pi^2} \int_{\mathbb{R}^2} \sum_{\mu,\nu} \mathbf{d}_\mu^{B+} \, T_{\mu\nu} \, w_\nu^A \, \mathbf{d}_\nu^{A+\mathsf{T}} \mathbf{M} \, \mathrm{e}^{\mathrm{i}(\mathbf{k}_\mu^{B+}\cdot\mathbf{x} - \mathbf{k}_\nu^{A+}\cdot\mathbf{x}')} \, \mathrm{d}^2 k, \tag{7}$$

the superscripts naming the medium whose wave numbers and polarisations are used. The blocks follow as in Eq. (3), with $\Lambda_a(\mathbf{k}_\mu^+)\,\mu_c(-\mathbf{k}_\nu^+)$ in place of $\Lambda_a\mu_c$ and the phase $\mathrm{e}^{\mathrm{i}(\mathbf{k}_\mu\cdot\mathbf{x}_m - \mathbf{k}_\nu\cdot\mathbf{x}_n)}$. The vertical phase of the reflected term is $\mathrm{e}^{\mathrm{i}(k_{z,\mu}|z| + k_{z,\nu}|z'|)}$: its decay on the evanescent branch is set by the distance of the receiver from the source's mirror image $\mathbf{x}'^* = (-z', x', y')$.

Where the plane waves fail. For cells in touching planes, $|z_m - z_n| = 2h$, the integral (3) does not converge: on the evanescent branch $k_{z,\nu} = \mathrm{i}\kappa$, $\kappa \to \infty$, the moments Λ_a and μ_c grow as $\mathrm{e}^{\kappa h}$ each and cancel the decay $\mathrm{e}^{-2\kappa h}$ of the phase. In the reflected coupling the relevant distance is that from the image, $|z_m + z_n|$, which is $2h$ for any two cells that touch the interface, and the same cancellation occurs (Figure 1a). The integrand then decays only algebraically in q, and so does the error of a truncated integral. Table 1 shows it for a cell that touches the interface, coupled to its own image: the contribution of each shell $Q < qh < 2Q$ falls by a factor of four from one shell to the next, so the error of an integral truncated at Q falls as Q^{-2}.

The singular part of the integrand is its behaviour at large q. There $k_{z,\nu} = \mathrm{i}q\,[1 - \omega^2/(2c_\nu^2 q^2) + O(q^{-4})]$, and the reflected and transmitted kernels tend to the static kernel of Section 3. At fixed

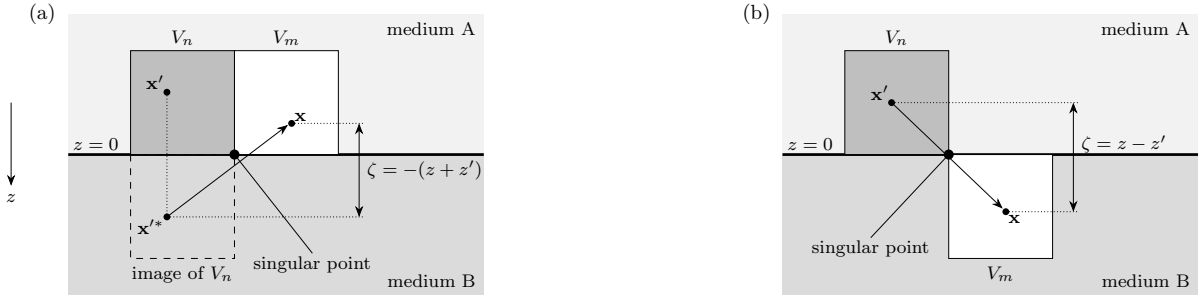


Figure 1. Cells at a welded interface, in a vertical section; z is depth. (a) Two cells in medium A that touch the interface and each other. The reflected coupling is the field of the source cell's mirror image (dashed), at the depth $\zeta = -(z + z')$ below the field point, and the image touches the field cell at the singular point. (b) A source cell in A and a field cell in B that touch at a point of the interface. The transmitted coupling depends on $\zeta = z - z'$, and the singular point is where the cells touch.

depths the difference is of relative order $\omega^2\zeta/(c^2q)$, since the correction to $k_{z,\nu}$ multiplies the depth ζ in the phase; in the block of cells that touch the interface the depths that matter are of order $1/q$, and the difference of the integrands is of relative order $(\omega/\beta q)^2$ (about $2(\omega/\beta q)^2$ for the cell of Table 1, from $qh = 20$ to 1280). Writing $\mathbf{K}^R = \mathbf{K}_0^R + (\mathbf{K}^R - \mathbf{K}_0^R)$, with $\mathbf{K}_0^R$ the block of the static image, the first term is obtained in closed form (Section 4) and the second by the plane-wave integral with the static spectrum subtracted under the integral sign. The remainder converges as Q^{-4} (Table 1): the subtraction gains exactly the two orders of $(\omega/\beta q)^2$, and at $Qh = 10$ it leaves 1.3×10^{-6} of the block where the truncated integral leaves 8×10^{-3}.

The kernels must be evaluated with care at large q. Written as a sum over the modes ν, as in Eqs. (6) and (7), the P and SV terms cancel as $(cq/\omega)^4$, because their bilinear-unit polarisations grow as cq/ω while their sum tends to the finite static kernel. In double precision the reflected kernel is then in error by 6×10^{-7} at $qh = 20$ for the cell of Table 1, and above $qh \approx 100$ the remainder after the subtraction is mostly rounding error. Table 1 was therefore computed with the kernels in 40-digit arithmetic, which agree with 60 digits to 3×10^{-16} at $qh = 640$. A production evaluation in double precision therefore needs a cutoff below $qh \approx 20$; Section 6 removes the next term of the expansion so that such a cutoff is accurate enough.

	reflected kernel		less its static part	
shell qh	contribution	order	contribution	order
10–20	5.8×10^{-3}		1.2×10^{-6}	
20–40	1.4×10^{-3}	2.02	7.7×10^{-8}	4.00
40–80	3.3×10^{-4}	2.13	4.3×10^{-9}	4.17
80–160	8.5×10^{-5}	1.95	2.8×10^{-10}	3.90
160–320	2.1×10^{-5}	2.01	1.8×10^{-11}	4.01
320–640	5.3×10^{-6}	2.00	1.1×10^{-12}	4.00

Table 1. The plane-wave integral of the reflected block of a cell touching the interface, coupled to its own image (field function 1, source monomial 1, media of Section 7, $\omega h/\beta = 0.17$): the contribution of each shell of the lateral wavenumber, relative to the block, and the local order $\log_2$ of the ratio of successive contributions. The kernels are evaluated in 40-digit arithmetic, the azimuthal integral by a trapezoid rule of $4qh + 64$ points and the radial one by Gauss rules on fixed panels; halving the panels and doubling the azimuthal rule changes the shell 160–320 by 6×10^{-20}.

3 The static field of a point force at a welded interface

In the lateral-wavenumber domain. Let a unit point force act in A at depth $z' < 0$. In the frame whose x axis lies along $\mathbf{k}$, the static Navier equations of an isotropic medium have the solutions $(a + bz)\,\mathrm{e}^{\pm qz}$, the double root replacing the separate P and S waves of the dynamic problem. The field in A is Kelvin's field plus a reflected field regular as $z \to -\infty$; the field in B is transmitted and regular as $z \to \infty$; displacement and traction on the planes $z = \text{const}$ are continuous at $z = 0$. This is the problem of Rongved [6], solved here in the lateral-wavenumber domain. Its solution has a simple structure: every entry of the reflected tensor is $\mathrm{e}^{q(z+z')}/q$ times a polynomial of degree at most one in each of qz and qz', and every entry of the transmitted tensor is $\mathrm{e}^{-q(z-z')}/q$ times a polynomial in qz and qz', the coefficients rational in the Lamé constants. When the two media coincide the reflected tensor vanishes and the transmitted one is Kelvin's.

In space. With $\zeta = -(z + z') > 0$ for the reflected field and $\zeta = z - z' > 0$ for the transmitted one, $\boldsymbol{\rho}$ the lateral separation and $R = (|\boldsymbol{\rho}|^2 + \zeta^2)^{1/2}$, the transform to space follows from

$$\frac{1}{4\pi^2}\int_{\mathbb{R}^2} \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\boldsymbol{\rho}}\, q^m \mathrm{e}^{-q\zeta}\,\mathrm{d}^2k = \begin{cases} (-\partial_\zeta)^{m+1}\Phi_0, & m \geq -1, \\ \Phi_1, & m = -2, \\ \Phi_2, & m = -3, \end{cases} \tag{8}$$

with the three potentials

$$\Phi_0 = \frac{1}{2\pi R}, \qquad \Phi_1 = -\frac{\log(R+\zeta)}{2\pi}, \qquad \Phi_2 = \frac{\zeta\log(R+\zeta) - R}{2\pi}, \tag{9}$$

each the ζ-antiderivative of the one before, $\partial_\zeta\Phi_1 = -\Phi_0$ and $\partial_\zeta\Phi_2 = -\Phi_1$. Rotating from the frame of $\mathbf{k}$ to fixed axes brings factors k_h/q; since $\mathrm{i}k_h$ becomes ∂_{ρ_h}, an entry $(k_h/q)f(q)$ is ∂_{ρ_h} of the transform of $f/(\mathrm{i}q)$, and an entry $(k_hk_l/q^2)f(q)$ is $-\partial_{\rho_h}\partial_{\rho_l}$ of the transform of f/q^2. These divisions by q are what produce the powers $m = -2$ and -3, and so the logarithmic potentials. They are the terms that Mindlin [7] found for the half-space with a free surface; at a welded interface they are present for every pair of media, with coefficients that vanish only when the media coincide.

Canonical terms. The point propagator differentiates the Green's tensor: a receiver derivative ∂_z acts on the explicit z and on ζ, as $\partial/\partial z - \partial_\zeta$ for the image and $\partial/\partial z + \partial_\zeta$ for the transmitted field, and similarly for source derivatives and the lateral ones. Collected, the 9×9 static image is a sum of 356 terms

$$c(\lambda_A, \mu_A, \lambda_B, \mu_B)\; z^A z'^B\; \partial_{\rho_x}^{\alpha_x}\partial_{\rho_y}^{\alpha_y}\partial_\zeta^{\alpha_z}\Phi_j(\boldsymbol{\rho}, \zeta), \tag{10}$$

in 61 families $(j, \boldsymbol{\alpha})$: Φ_0 with up to four derivatives, and Φ_1 and Φ_2 under lateral derivatives only (one to four, and two to four), since a ζ-derivative of a logarithmic potential is the potential below it. The transmitted field has 275 terms in 46 families of the same kind. Every kernel $\partial^\alpha\Phi_j$ is homogeneous in $(\boldsymbol{\rho}, \zeta)$, of degree $-1 - |\boldsymbol{\alpha}|$, $-|\boldsymbol{\alpha}|$ and $1 - |\boldsymbol{\alpha}|$ for $j = 0, 1, 2$.

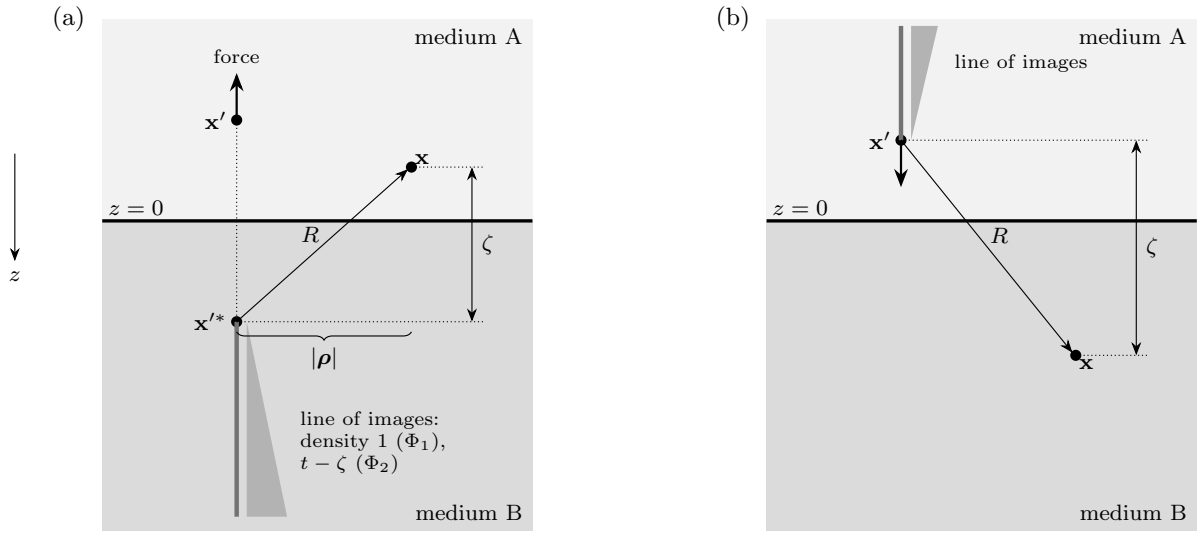


Figure 2. The image sources of the static fields at a welded interface, in a vertical section; z is depth. (a) Reflection: a point force at $\mathbf{x}'$ in medium A is seen from a receiver $\mathbf{x}$ in A through its Kelvin image at the mirror point $\mathbf{x}'^*$, at the distance R and the depth $\zeta = -(z + z')$ below the receiver, and through Mindlin's terms, a line of images on the vertical half-line beyond $\mathbf{x}'^*$ with uniform density (Φ_1) and density growing as $t - \zeta$ (Φ_2), Eq. (11). (b) Transmission: the field in B has no image point; its potentials are centred on the source itself, with $\zeta = z - z'$, and its line of images lies on the vertical half-line above $\mathbf{x}'$.

A line of images. Every logarithmic term carries a lateral derivative and therefore vanishes as $\zeta \to \infty$. Integrating the relations between the potentials from ζ to infinity gives

$$\partial^\alpha \Phi_1(\boldsymbol{\rho}, \zeta) = \int_\zeta^\infty \partial^\alpha \Phi_0(\boldsymbol{\rho}, t)\, \mathrm{d}t, \qquad \partial^\alpha \Phi_2(\boldsymbol{\rho}, \zeta) = \int_\zeta^\infty (t - \zeta)\, \partial^\alpha \Phi_0(\boldsymbol{\rho}, t)\, \mathrm{d}t. \tag{11}$$

Mindlin's terms are thus Kelvin's image spread along the vertical half-line beyond the image point, with uniform and with linearly growing density (Figure 2). This is the form in which their moments are computed.

4 Coupling blocks with the static image

The separation form. For two cells in A the image kernel depends on the lateral differences $\rho_h = x_h - x'_h$ but on the vertical sum $\zeta = -(z + z')$. The block is therefore an integral over (ρ_x, ρ_y, ζ) against a weight that is a cross-correlation of the cells' factors in the two lateral coordinates, as in the homogeneous medium, and a convolution in the vertical, $u_{\alpha\beta}(\tau) = \int v^\alpha (\tau - v)^\beta \, \mathrm{d}v$ in $\tau = v + v'$, into which the factors z^A and z'^B of Eq. (10) are expanded. The weights are piecewise polynomials with rational coefficients, computed exactly. The domain divides into boxes on which the weight is a polynomial.

The singular point. The image kernel is singular only at $\boldsymbol{\rho} = 0$, $\zeta = 0$, which is reached when both cells touch the interface and touch each other laterally (Figure 1a), and then at a vertex of the boxes that hold it. There the vertical weight vanishes as ζ^{1+A+B}, and for every one of the

356 terms the integrand is at worst $O(R^{-2})$: it is absolutely integrable, with no distribution and no derivative to be moved onto the weight. Boxes away from the singular point have smooth integrands and are integrated by Gauss rules in coordinates centred on each box. On a box with the singular point at a vertex, reflected onto the positive octant, every term requires the corner moments

$$M_{j,\alpha}[p,q,r] = \int_0^a\int_0^b\int_0^c \rho_x^p\rho_y^q\zeta^r\,\partial^\alpha\Phi_j(\boldsymbol{\rho},\zeta)\,\mathrm{d}\zeta\,\mathrm{d}\rho_y\,\mathrm{d}\rho_x, \tag{12}$$

with $a = b = c = 2h$. They are universal: by homogeneity they scale as a power of h times a number independent of h, the frequency and the media.

Kelvin's potential. For $j = 0$, $\partial^\alpha R^{-1} = P_\alpha(\boldsymbol{\rho},\zeta)\,R^{-2|\alpha|-1}$ with P_α a homogeneous polynomial of degree $|\boldsymbol{\alpha}|$, so the moments are combinations of the master integrals

$$I[p,q,r;n] = \int_0^a\int_0^b\int_0^c x^p y^q z^r\,(x^2+y^2+z^2)^{n/2}\,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x, \tag{13}$$

here for odd n down to -9. The integrand is homogeneous of degree $p+q+r+n$ about the singular point, and Euler's identity with the divergence theorem reduces the box to its three far faces, the faces through the singular point carrying no flux; on a face the same step in two dimensions reduces to edges, and on an edge the integral is elementary [1]. One transcendental function enters, through the face integral of R^{-3}: $\int_0^b\int_0^c R^{-3}\,\mathrm{d}z\,\mathrm{d}y = a^{-1}\arctan\big(bc/(a\sqrt{a^2+b^2+c^2})\big)$ on the face $x = a$, which is continued to lower n by differentiation with respect to a.

The logarithmic potentials. By Eq. (11) the ζ-integral of the box can be done first, which leaves Kelvin's potential against a polynomial weight in t over the box and over the semi-infinite column $[0,a]\times[0,b]\times[c,\infty)$ above it. On $[0,c]$ the weight is $t^{r+1}/(r+1)$ ($j=1$) or $t^{r+2}/((r+1)(r+2))$ ($j=2$); on the column it is $c^{r+1}/(r+1)$, or $t\,c^{r+1}/(r+1) - c^{r+2}/(r+2)$. The column is the semi-infinite box less the finite one. Euler's identity holds on the semi-infinite box, the face at infinity carrying no flux, and its faces reduce to edges that run to infinity,

$$\int_0^\infty z^r(A^2+z^2)^{n/2}\,\mathrm{d}z = \tfrac{1}{2}\,A^{r+n+1}\,\mathrm{B}\Big(\tfrac{r+1}{2},-\tfrac{r+n+1}{2}\Big), \tag{14}$$

Beta functions at half-integers, elementary. Two cases need care. Where the degree $p+q+r+n+3$ of a column term is negative, each of the two boxes holds an integral that diverges at the singular corner; Euler's identity then gives Hadamard's finite part of each, and the two finite parts differ by the exact integral over the column, because their singular contributions at the shared corner are equal. Where the degree is zero the identity is degenerate, and the column is reduced instead by integration by parts in t, then in ρ_x and ρ_y, whose boundary terms are faces at height c and strips at $\rho_x = a$ or $\rho_y = b$, away from the singular point. Two base integrals remain:

$$B_1 = \int_0^a\int_0^b\int_c^\infty R^{-3}\,\mathrm{d}t\,\mathrm{d}\rho_y\,\mathrm{d}\rho_x, \qquad B_2 = \int_0^a\int_0^b\int_c^\infty \rho_x\rho_y\,R^{-5}\,\mathrm{d}t\,\mathrm{d}\rho_y\,\mathrm{d}\rho_x. \tag{15}$$

The base integrals. Both are of degree zero, so they depend only on the ratios of the sides, which here are equal. The t-integrals are elementary, and in polar coordinates (r,θ) in the ρ-plane so are the radial ones: with $c = 1$ and $s = \sqrt{1+r^2}$,

$$B_1 = \int \log\frac{1+s}{2}\,\mathrm{d}\theta, \qquad B_2 = \int \cos\theta\sin\theta\Big[\frac{1}{3s} + \frac{2}{3}\log(1+s) - \frac{1}{3} - \frac{2}{3}\log 2\Big]\mathrm{d}\theta,$$

with $r = \sec\theta$ on $[0,\pi/4]$ and $r = \csc\theta$ on $[\pi/4,\pi/2]$, the two halves equal by symmetry. For B_2 the substitution $v = \cos^2\theta$ makes the remaining integral elementary. For B_1 the integral $2\int_0^{\pi/4}\log\big(\frac{1}{2}(1+\sqrt{1+\sec^2\theta})\big)\mathrm{d}\theta$ evaluates to Catalan's constant G, $\pi\log 2$, and the imaginary parts of $\mathrm{Li}_2(2^{-1/2}\mathrm{e}^{\mathrm{i}\pi/12})$ and $\mathrm{Li}_2(2^{-1/2}\mathrm{e}^{7\mathrm{i}\pi/12})$. Kummer's formula writes $\mathrm{Im}\,\mathrm{Li}_2(r\mathrm{e}^{\mathrm{i}\theta})$ in Clausen functions $\mathrm{Cl}_2(\theta) = \mathrm{Im}\,\mathrm{Li}_2(\mathrm{e}^{\mathrm{i}\theta})$ through the auxiliary angle $\omega = \arctan\big(r\sin\theta/(1-r\cos\theta)\big)$ [8], and at both points $\omega = \pi/6$. The logarithms then cancel, and with $\mathrm{Cl}_2(\pi/2) = G$ and the duplication formula $\mathrm{Cl}_2(2\theta) = 2\mathrm{Cl}_2(\theta) - 2\mathrm{Cl}_2(\pi-\theta)$ at $\theta = \pi/6$,

$$B_1 = \tfrac{5}{4}\,\mathrm{Cl}_2\big(\tfrac{\pi}{3}\big) - G = 0.35271\,14138\ldots, \qquad B_2 = \tfrac{1}{3}\log\frac{(1+\sqrt{2})^2}{2(1+\sqrt{3})} = 0.02151\,58182\ldots. \tag{16}$$

Every corner moment is therefore elementary apart from the two constants G and $\mathrm{Cl}_2(\pi/3)$. All of them are universal numbers, computed once and stored.

A well-conditioned evaluation. The corner moments are monomial moments about the singular corner, and contracting them with the weights expanded in powers about that corner is ill-conditioned in the way the power moments of a homogeneous medium are [1]: at degree two it loses about a digit (Table 4). The blocks are therefore evaluated in a well-conditioned basis, and the corner moments serve as an independent check. On a box away from the singular point the weights are written, exactly, in Legendre polynomials of the box's own coordinates and contracted with the Legendre moments of the kernel, which a tensor Gauss rule gives. A box anchored at the singular point is split into three Duffy pyramids about it [9]. In pyramid k, with $t_k = \rho$ and $t_j = \rho v_j$ for the other two axes in units of the box's sides, every kernel $\partial^\alpha\Phi_j$ is ρ^d times its value on the far face, d its degree of homogeneity, and each one-dimensional weight is $(A_i\rho w_i)^{m_i} r_i(A_i\rho w_i)$, its zero of order m_i at the corner factored out exactly and r_i written in Legendre polynomials. The radial integrand is then $\rho^{2+d+\sum m_i}$ times a polynomial, integrated exactly, and the face integrand is smooth, the far face lying at a distance of two half-widths from the singular point. Two details decide the last digit. The Gauss weights are taken from a 30-digit table, since those of standard library routines carry relative errors growing with the order, of 4×10^{-15} at 32 points and up to 7×10^{-13} at 40; and the tensor-product sums are carried out one axis at a time, since a single running sum over the 32^3 nodes of a box loses about 2×10^{-15}.

5 Cells on opposite sides of the interface

For a source cell in A and a field cell in B the static coupling is the transmitted field, the whole of it, with no direct part. Its potentials are those of Eq. (9) with $\zeta = z - z' = |z| + |z'|$, and the singular point is where the two cells touch at the interface (Figure 1b). The 275 canonical terms

are integrable at that point, as the image's are. In the separation form the vertical weight is now a cross-correlation in $v - v'$, like the lateral ones, and the corner moments are those of Section 4.

The other arrangements follow by symmetry. A field cell in A with a source in B, and two cells in B, are the problems above reflected in the plane $z = 0$ with the two media exchanged. Under the reflection the displacement u_z and the shear strains γ_{zx}, γ_{zy} change sign, so the block is conjugated by $\mathrm{diag}(-1,1,1,1,1,1,1,-1,-1)$ in the order $(u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, \gamma_{xy}, \gamma_{zx}, \gamma_{zy})$, and each cell polynomial is multiplied by (-1) to its power of ξ_z.

6 The next term: the order ω^2

The static image removes the part of the reflected integrand that does not decay. What remains decays as the square of the frequency over the wavenumber, and its leading term can be removed the same way. Let the reflected displacement tensor of Section 3, in the frame of $\mathbf{k}$, be expanded at fixed (q, z, z') in powers of the frequency,

$$\mathbf{G}^R(q; z, z'; \omega) = \mathbf{G}_0^R + \omega^2\, \mathbf{G}_1^R + O(\omega^4), \tag{17}$$

with $\mathbf{G}_0^R$ the static image. With $\mathbf{u} = \mathbf{u}_0 + \omega^2 \mathbf{u}_1 + \dots$ the Navier equations $\nabla \cdot \boldsymbol{\sigma}(\mathbf{u}) + \rho\, \omega^2 \mathbf{u} + \mathbf{F}\delta = \mathbf{0}$ give, at order ω^2,

$$\nabla \cdot \boldsymbol{\sigma}(\mathbf{u}_1) = -\rho\, \mathbf{u}_0, \tag{18}$$

with no source: $\mathbf{u}_1$ and its traction are continuous at $z = z'$ and at $z = 0$. In each region the static field $\mathbf{u}_0$ is a polynomial times $\mathrm{e}^{\pm qz}$, the exponential being a double root of the static operator, so a particular solution of Eq. (18) is a polynomial of degree two higher times the same exponential. To it are added the static solutions $(a + bz)\mathrm{e}^{\pm qz}$ regular in the region. The whole-space problem is solved first; its ω^2 term, that of Kelvin's field, then forces the interface conditions that fix the reflected term $\mathbf{G}_1^R$ and the transmitted term $\mathbf{G}_1^T$. Every entry of $\mathbf{G}_1^R$ is $\mathrm{e}^{q(z+z')}$ times a polynomial in z and z', of degree at most two in each and three in all, with coefficients rational in q and in the Lamé constants and densities of the two media. The point propagator follows from $\mathbf{G}_1^R$ by the derivatives of Section 3, and its vertical moments over a cell are elementary.

Three checks confirm the result. When the media coincide, $\mathbf{G}_1^R = \mathbf{0}$ and $\mathbf{G}_1^T$ is the ω^2 term of Kelvin's field. Against the exact kernels, evaluated in 50-digit arithmetic, the residual $\mathbf{G} - \mathbf{G}_0 - \omega^2 \mathbf{G}_1$ falls sixteenfold each time ω is halved, as an $O(\omega^4)$ remainder must. And the derivation done independently in two computer-algebra systems, one with symbolic and one with rational material constants, agrees to 5×10^{-51}.

In space the term is not needed, and does not come cheaply. The divisions by q that turn a rotated entry into a lateral derivative (Section 3) take the lowest power of q in $\mathbf{G}_1^R$ to $q^{-5}\mathrm{e}^{-q\zeta}$, whose transform, like that of $q^{-4}\mathrm{e}^{-q\zeta}$, exists only as a finite part: two potentials beyond Φ_2 would be needed, and the moments of their finite parts. We use the term where it is simple, in the wavenumber domain. With $\mathbf{Y}(\mathbf{k}; \omega)$ the integrand of the reflected block, so that

$\mathbf{K}^R = (4\pi^2)^{-1}\int \mathbf{Y}\,\mathrm{d}^2k$, and $\mathbf{Y}_0$, $\mathbf{Y}_1$ the integrands built in the same way from $\mathbf{G}_0^R$ and $\mathbf{G}_1^R$,

$$\mathbf{K}^R = \mathbf{K}_0^R + \omega^2\,\mathbf{K}_1^R(Q_0) + \frac{1}{4\pi^2}\int_{\mathbb{R}^2}\left[\mathbf{Y} - \mathbf{Y}_0 - \omega^2\,\mathbf{Y}_1\,\theta(q-Q_0)\right]\mathrm{d}^2k, \qquad \mathbf{K}_1^R(Q_0) = \frac{1}{4\pi^2}\int_{q>Q_0}\mathbf{Y}_1\,\mathrm{d}^2k. \tag{19}$$

Neither $\mathbf{K}_0^R$ nor $\mathbf{K}_1^R(Q_0)$ depends on the frequency: both are computed once for each arrangement of cells, the first in closed form (Section 4) and the second as a plane-wave integral of a closed-form integrand, which needs no reflection coefficients. Only the last integral is evaluated at each frequency, and beyond Q_0 its integrand falls by two more powers of q than with the static image alone.

Table 2 shows the effect for the block of Table 1, at $\omega h/\beta = 1/6$ and at $\omega h/\beta = 1$, the frequency at which a shear wavelength spans about three cells. The remainder now converges as Q^{-6}. Truncated at $Qh = 10$ it leaves 3.6×10^{-10} of the block at the lower frequency, where the static subtraction alone leaves 1.3×10^{-6} and needs $Qh \approx 160$ to do as well; at $\omega h/\beta = 1$ it leaves 3×10^{-7} at $Qh = 10$ and 6×10^{-11} at $Qh = 40$, where the static subtraction alone leaves 3×10^{-5} and 1×10^{-7}. The transmitted coupling behaves in the same way. For a source cell touching the interface and the field cell its mirror image in B, so that the two touch across the interface, the shells of $\mathbf{Y} - \mathbf{Y}_0$ fall with local orders 3.90 to 4.17 and those of $\mathbf{Y} - \mathbf{Y}_0 - \omega^2\mathbf{Y}_1$ (with $\mathbf{Y}_1$ built from $\mathbf{G}_1^T$) with orders 5.86 to 6.24; truncated at $Qh = 10$ the remainder leaves 2.2×10^{-10} of the block at $\omega h/\beta = 1/6$, and truncated at $Qh = 40$ it leaves 4.3×10^{-11} at $\omega h/\beta = 1$.

The lower cutoff also settles the precision of the mode sums. Below $Qh = 10$ the cancellation between the P and SV terms (Section 2) is mild: evaluated in double precision, the integral over $qh < 10$ differs from its 40-digit value by 7×10^{-12} of the block at $\omega h/\beta = 1/6$ and by 5×10^{-13} at $\omega h/\beta = 1$, and the shell $10 < qh < 20$ by 6×10^{-11} and 2×10^{-14}. With the ω^2 term subtracted, a double-precision plane-wave integral therefore gives the reflected block of a touching cell to about 4×10^{-10} with the cutoff at $Qh = 10$ when $\omega h/\beta = 1/6$, and to about 10^{-10} with the cutoff at $Qh = 40$ when $\omega h/\beta = 1$; with the static image alone the same accuracy needs cutoffs at which double precision fails. The frequency-independent integral $\mathbf{K}_1^R(Q_0)$ is the slowly convergent one, with an integrand that falls as $\mathbf{Y} - \mathbf{Y}_0$ does. Its integrand needs no mode sum: written in the frame of $\mathbf{k}$ it is $\mathrm{e}^{q(z+z')}$ times a polynomial in z and z' whose coefficients are Laurent polynomials in q, and in double precision it agrees with the 40-digit kernel to 10^{-12}, an error that enters multiplied by ω^2. It is integrated once, to $Q_1h = 160$, and the tail beyond Q_1 is taken from the q^{-4} law of the last octave. The angular dependence costs little: the rotation of the frame kernel to the direction of $\mathbf{k}$ contains only the harmonics $\mathrm{e}^{\mathrm{i}n\phi}$ with $|n| \le 4$, so at each q the azimuthal rule only sums the cells' lateral moments against nine harmonics.

Assembled in this way, with the plane-wave integral below Q_0 in double precision, the split reproduces the reference values to the size of the remainder it drops. For identical media the transmitted block must be the touching block of the homogeneous medium, which Nestor [1] obtains in closed form with no plane waves; over every field function and source monomial ($4\times10\times9\times9$ entries), for cells touching across the interface face to face, along an edge and at a corner, and with the source above or below, the two agree to 0.8 to 3.2×10^{-10} at $\omega h/\beta = 1/6$ with $Q_0h = 10$ and to 1.7×10^{-12} to 5.4×10^{-11} at $\omega h/\beta = 1$ with $Q_0h = 40$. For different media the reflected block of a cell with its own image and the transmitted block of two cells

touching across the interface agree with the 40-digit integrals of Tables 1 and 2 to 3.6×10^{-10} and 2.2×10^{-10} at the lower frequency and to 5.9×10^{-11} and 3.9×10^{-11} at the higher, the remainders of Table 2 and of the transmitted sweep.

	$\omega h/\beta = 1/6$				$\omega h/\beta = 1$			
	less $\mathbf{Y}_0$		less $\mathbf{Y}_0 + \omega^2\mathbf{Y}_1$		less $\mathbf{Y}_0$		less $\mathbf{Y}_0 + \omega^2\mathbf{Y}_1$	
shell qh	contr.	order	contr.	order	contr.	order	contr.	order
10–20	1.2×10^{-6}		3.5×10^{-10}		3.0×10^{-5}		3.1×10^{-7}	
20–40	7.7×10^{-8}	4.00	5.5×10^{-12}	6.00	1.8×10^{-6}	4.01	4.7×10^{-9}	6.01
40–80	4.3×10^{-9}	4.17	7.2×10^{-14}	6.25	1.0×10^{-7}	4.18	6.2×10^{-11}	6.25
80–160	2.8×10^{-10}	3.90	1.2×10^{-15}	5.87	6.8×10^{-9}	3.90	1.1×10^{-12}	5.87
160–320	1.8×10^{-11}	4.01	1.9×10^{-17}	6.03	4.2×10^{-10}	4.01	1.6×10^{-14}	6.02

Table 2. The block of Table 1 with the static image subtracted, and with the static image and the ω^2 term subtracted, at two frequencies: the contribution of each shell of the lateral wavenumber, relative to the block, and the local order. All kernels in 40-digit arithmetic; halving the radial panels and doubling the azimuthal rule changes the shell 160–320 of the second remainder by 8×10^{-20} and 5×10^{-20} of the block.

7 Verification

Each step was checked against a route that shares no step with it; Table 3 collects the results. The media are A with $\alpha = 5.0$, $\beta = 3.0\,\mathrm{km\,s^{-1}}$, $\rho = 2.5\,\mathrm{g\,cm^{-3}}$ and B with $\alpha = 6.5$, $\beta = 3.6\,\mathrm{km\,s^{-1}}$, $\rho = 2.8\,\mathrm{g\,cm^{-3}}$, and $h = 0.05\,\mathrm{km}$. Errors are relative to the largest entry of the quantity compared, and the corner moments are measured on the scale 2^{p+q+r} of each monomial.

The checks are of four kinds. The spectral solutions were derived independently in two computer-algebra systems, and their spatial forms were compared with direct two-dimensional Fourier integration of the spectral kernels. The corner moments were compared with three routes: Euler's reduction with Gauss rules on the far faces, tanh-sinh quadrature of the singular volume integral itself, and an evaluation at thirty digits that integrates one coordinate exactly and then the other two with Duffy coordinates at the corner [9]. We note that adaptive three-dimensional cubature applied directly with the singular point at a corner, or over the semi-infinite column, can report convergence while in error by as much as 5×10^{-7}, which is why the references integrate one coordinate exactly. The blocks of cells away from the singular point were compared with a tensor Gauss rule over both cells of the canonical kernel, which shares neither the separation form nor the exact weights. Two limits close the chain: for identical media the transmitted blocks must be the touching blocks of the homogeneous medium [1], computed by an unrelated method, and the image must vanish; and the touching block of a cell with its own image must be the limit of the plane-wave integral (6) of the static spectrum as the cutoff grows.

The accuracy of the touching blocks themselves is measured against a reference that removes the rounding of the double-precision routes: the power-basis route carried out in 80-bit extended arithmetic (unit round-off 1.1×10^{-19}), with the corner moments in closed form at 40 digits, Gauss rules refined to extended precision, and the kernels and exact weights rounded to extended precision only. Its own rounding, bounded by the unit round-off times the sum of the magnitudes

of its terms, is below 6×10^{-17} of the blocks. Against it (Table 4) the well-conditioned evaluation is at round-off for both field degrees, 3×10^{-16} to 1.2×10^{-15}, and the power basis is the same at degree one but loses up to a digit at degree two. The fully independent check of these blocks, the plane-wave integral, is limited to 2.3×10^{-8} by its own convergence. With identical media the image vanishes identically, to the last bit, on both sides of the interface.

quantity	compared with	agreement
Plane-wave representation		
integrand of Eq. (2)	residue form of the plane-to-plane kernel	3×10^{-15}
blocks (3), planes not touching	closed-form blocks [1]	2–6×10^{-13}
Static image		
reflected spectral tensor	second computer-algebra derivation	5×10^{-16}
dynamic reflected integrand of a touching block, $qh = 20$–1280	static integrand	$\approx 2(\omega/\beta q)^2$
spatial form	Fourier integral of the spectrum	7×10^{-13}–1.3×10^{-12}
canonical terms (10)	direct differentiation of the spatial kernel	3×10^{-15}
Corner moments		
Euler's reduction, 7 cases, up to 4 derivatives	tanh-sinh quadrature of the volume integral	6×10^{-16}–4.8×10^{-14}
closed forms, all 53 352 moments, monomials to degree $(6, 6, 8)$	Euler's reduction with Gauss rules on the faces	1.2×10^{-15}
13 moments, all potentials, 3 degenerate columns	thirty-digit evaluation	5×10^{-23}–1.6×10^{-16}
B_1, B_2, Eq. (16)	forty-digit angular quadrature	10^{-40}
Reflected blocks		
cells away from the singular point, 4 pairs, degree-one and degree-two fields	tensor Gauss rule over both cells	3×10^{-15}–8.5×10^{-15}
touching blocks, closed-form corner moments, 3 pairs	Gauss rules on the far faces	1.8–5.3×10^{-15}
cell with its own image	static plane-wave integral, $Qh = 320$ to $20\,480$	$7.9 \times 10^{-6} \to 2.3 \times 10^{-8}$
Transmitted field and blocks		
transmitted spectral tensor	second computer-algebra derivation	4.2×10^{-16}
spatial form	Fourier integral of the spectrum	3.9×10^{-16}
canonical terms, different media	Fourier integral of the spectrum	6.6–7.8×10^{-13}
identical media: face, edge, corner contact; mirrored; degree two	homogeneous-medium blocks [1]	1.9–7.6×10^{-15}
different media, cells not touching	tensor Gauss rule over both cells	1.7–5.0×10^{-13}
Order ω^2 and the split (19)		
$\mathbf{G}_1^R$, $\mathbf{G}_1^T$	exact kernels at 50 digits, ω halved	residual falls 16-fold
$\mathbf{G}_1^R$, $\mathbf{G}_1^T$	second computer-algebra derivation	5×10^{-51}
frame integrands $\mathbf{Y}_0$, $\mathbf{Y}_1$ in double precision, $qh = 0.5$–160	40-digit kernels	2×10^{-14}, 1.2×10^{-12}
identical media, touching across the interface, 3 contacts, mirrored	homogeneous-medium blocks [1]	1.7×10^{-12}–3.2×10^{-10}
different media, reflected and transmitted touching blocks	40-digit plane-wave integrals	3.9×10^{-11}–3.6×10^{-10}

Table 3. Independent checks. The last entry of each group of blocks is limited by the accuracy of the comparison route: the Gauss rule over both cells at order 10 to about 10^{-13}, and the static plane-wave integral by its algebraic convergence; those of the split are the remainder it drops, at $Q_0 h = 10$ ($\omega h/\beta = 1/6$) and $Q_0 h = 40$ ($\omega h/\beta = 1$).

cells	field	power basis	well-conditioned
reflected, cell with its own image	degree one	9.2×10^{-16}	8.2×10^{-16}
reflected, lateral neighbour	degree one	6.8×10^{-16}	7.3×10^{-16}
reflected, diagonal neighbour	degree one	6.6×10^{-16}	4.0×10^{-16}
transmitted, face contact	degree one	1.2×10^{-15}	1.2×10^{-15}
transmitted, corner contact	degree one	3.4×10^{-16}	4.0×10^{-16}
reflected, cell with its own image	degree two	9.6×10^{-15}	9.2×10^{-16}
reflected, diagonal neighbour	degree two	1.7×10^{-15}	6.5×10^{-16}
transmitted, face contact	degree two	9.6×10^{-15}	4.1×10^{-16}

Table 4. The touching blocks against the extended-precision reference: for each kernel family, the largest error over the field and source functions relative to the largest entry, and the largest over the families. Degree one: 4 field functions and 10 source monomials; degree two: 10 and 35.

8 Discussion

The static image carries the whole singularity of the coupling at a welded interface, and in closed form it costs no more than the touching blocks of a homogeneous medium: a fixed set of universal corner moments, contracted with exact rational weights and with coefficients rational in the Lamé constants. The dynamic remainder is smooth, and the plane waves of Section 2 carry it at a cost that does not depend on how close the cells are to the interface, since the subtraction has removed what made the integral converge slowly.

The split also preserves what makes a frequency sweep cheap in a homogeneous medium, where the closed-form blocks are set up once and each frequency costs little. Here the static blocks $\mathbf{K}_0^R$ and the coefficients $\mathbf{K}_1^R(Q_0)$ of the ω^2 term are computed once for each arrangement of cells, and at each frequency only the remainder is integrated, over $qh < Q_0 h$, with reflection coefficients evaluated on a grid of lateral wavenumbers that every block shares. Terms of order ω^4 and beyond could be derived the same way, each a polynomial two degrees higher; but with the cutoff already where double precision holds, the next gain would be in the cost of the remainder, not in its accuracy.

For a stack of layers the same construction applies interface by interface. The singular part of the reflected and transmitted couplings comes only from the interface that the cells touch, and is the static image of that interface; reflections from other interfaces reach a cell after travelling at least one layer thickness and are smooth. The plane-wave representation extends to the stack through the reflection and transmission matrices of the layered medium, composed by the recursions of the reflectivity method [2, 3]. The coupling of every cell with every other is then available from the up- and down-going factorisation alone, without tabulating the stratified Green's tensor, which is the storage advantage that motivates the approach; a two-way march in depth that applies the volume operator in this way is the subject of further work.

One limitation remains: the closed form of the base integrals is stated for cubic corner boxes, $a = b = c$, which is all a grid of cubes requires; for unequal sides the angular integrals are evaluated by quadrature.

Data availability

The Mathematica notebooks and Python scripts that derive and check every result, and the package they test, are available at https://github.com/tmnestor/MultipleScatteringCalculations: the static image and transmitted field, `Mathematica/StaticInterfaceImage.wl`, `Mathematica/StaticImageSpace.wl` and `Mathematica/StaticInterfaceTransmission.wl`, with their Python twins `scripts/derive_static_interface_image.py`, `scripts/derive_static_image_space.py` and `scripts/derive_static_interface_transmission.py`; the canonical terms, `scripts/derive_interface_image_terms.py`; the corner moments and base integrals, `Mathematica/ImageCornerMoments.wl`, `scripts/derive_image_corner_moments.py` and `cubic_scattering/image_moments.py`; the blocks, `cubic_scattering/interface_image.py`; the plane-wave representation, reflection and transmission, `cubic_scattering/`

`stratified_march.py`; the checks of Table 3, `scripts/gate_march_modes.py` and `scripts/gate_interface_image.py`; the term of order ω^2, `Mathematica/InterfaceOmega2.wl` and `scripts/derive_interface_omega2.py`; and the convergence studies of Tables 1 and 2, `scripts/measure_interface_static_subtraction.py` and `scripts/measure_interface_omega2_subtraction.py`; the split at many frequencies, `cubic_scattering/interface_spectral.py`, checked by `scripts/gate_interface_spectral.py`.

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This work began thirty years ago in the author's doctoral research at the Research School of Earth Sciences of the Australian National University, Canberra [4].

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

Software development. Software development followed a human-directed, AI-assisted workflow, organised as a Test Driven Hypothesis Acceptance cycle: each derivation, implementation or numerical claim was stated as a hypothesis together with a test designed to refute it (an independent route, an exact limit, a high-precision reference or a convergence study), and was accepted only when the test passed; a failed test returned the hypothesis for diagnosis, never the test for adjustment. The author designed the software originally, but only directed and validated recent improvements, which build on the author's original Fortran implementation (the `PhD_fortran_code` folder of the repository given under Data availability) and an extensive library of Mathematica notebooks, developed and extended since 1992, that carry out and verify the symbolic derivations; an agentic coding assistant (Anthropic Claude) operated under human-in-the-loop supervision to implement specified changes and diagnostic tests. The design, the choice of numerical methods, and the resolution of the principal technical difficulties were the author's. All AI-generated changes were subject to code review by the author and validated by automated tests and independent checks.

Manuscript preparation. During the preparation of this work, the author used Claude (Anthropic) in order to search for relevant open-access research, to write the TikZ code of the diagrams and to help redraft the paper into more coherent sections for readability. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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